Hints: Polynomials – 9. Cyclotomic polynomials and the arithmetic of roots of unity

Class work

The hints to each problem are cumulative. Read one, try again, and open the next only if you are still stuck.

Problem 1

Hint 1. Follow the proof idea of Theorem 2: if f is the minimal polynomial of ζ, $\Phi_n = fg$ and $g(\zeta^p) = 0$ for a prime $p \nmid n$, then $f(x) \mid g(x^p) \equiv g(x)^p \pmod p$, so $\bar f$ and $\bar g$ share a factor and $x^n - 1$ is not squarefree over $\mathbb{F}_p$, contradicting $\gcd(x^n - 1, nx^{n-1}) = 1$. So $f(\zeta^p) = 0$.

Hint 2. The argument applies to every root of f, so you can iterate: for k coprime to n write $k = p_1 \cdots p_s$ and pass $\zeta \to \zeta^{p_1} \to \zeta^{p_1 p_2} \to \cdots$; each step stays a primitive n-th root and a root of f. Hence all primitive roots are roots of f, and $f = \Phi_n$.

Hint 3. $\Phi_n(1)$: put $x = 1$ in $\frac{x^n-1}{x-1} = 1+x+\cdots+x^{n-1} = \prod_{d\mid n,\, d>1} \Phi_d(x)$ to get $n = \prod_{d\mid n, d>1} \Phi_d(1)$, and induct on n (for $n = p^a$ the factors are $\Phi_{p^i}(1)$; otherwise the prime powers dividing n already use up all of n). Sum of primitive roots $s(n)$: $\sum_{d\mid n} s(d)$ is the sum of all n-th roots of unity, which is $[n = 1]$, so $s = \mu$ by Möbius inversion.

Problem 2

Hint 1. By CRT write $\zeta^k = \omega^i \eta^j$ with $\omega = e^{2\pi i/p}$, $\eta = e^{2\pi i/q}$ and $k \leftrightarrow (i, j) \in \mathbb{Z}/p \times \mathbb{Z}/q$; the relation becomes $\sum_j (\sum_i a_{ij}\omega^i)\eta^j = 0$.

Hint 2. Since $[\mathbb{Q}(\zeta) : \mathbb{Q}] = (p-1)(q-1)$, Φ_q stays irreducible over $\mathbb{Q}(\omega)$, so the only linear relation among $1, \eta, \ldots, \eta^{q-1}$ over $\mathbb{Q}(\omega)$ is "all coefficients equal". Hence $c_j = \sum_i a_{ij}\omega^i$ does not depend on j, so for fixed j, j' the numbers $a_{ij} - a_{ij'}$ do not depend on i.

Hint 3. So $a_{ij} = b_i + t_j$; normalizing $\min_i b_i = 0$ makes all b_i, t_j non-negative integers. The term t_j (for all i) is a regular p-gon, the term b_i (for all j) a regular q-gon. Lam–Leung for pq: the number of roots is $p\cdot(\#p\text{-gons}) + q\cdot(\#q\text{-gons})$. For $n = 30$: $\sum_{k=1}^4 e^{2\pi i k/5} = -1 = -(e^{\pi i/3} + e^{5\pi i/3})$, and one checks that no 2, 3 or 5 of the six numbers form a regular polygon.

Problem 3

Hint 1. Equal angles mean the side directions are ε^j with $\varepsilon = e^{2\pi i/pq}$, so the side lengths a_j satisfy $\sum_j a_j \varepsilon^j = 0$. Apply Problem 2 (its proof works for any rational, in particular integer, coefficients).

Hint 2. The proof of Problem 2 gives $a_j = u_{j \bmod p} + v_{j \bmod q}$ for suitable numbers u_r, v_s. For $k \leqslant p < q$ consider the k^2 numbers $u_r + v_s$ with $r \in \{1, \ldots, k\} \bmod p$, $s \in \{1, \ldots, k\} \bmod q$.

Hint 3. By CRT each such sum is the length of some side, and these are k^2 different sides, so their lengths are distinct positive integers with total $\geqslant 1 + 2 + \cdots + k^2 = \frac{k^2(k^2+1)}{2}$. But this total equals $k\sum_{i\leqslant k}(u_i + v_i) = k\sum_{i=1}^k a_i$. Divide by k.

Problem 4

Hint 1. Write $a_i = i\,(i^{k-1} + 1)$. Use two consequences of the a_i being a permutation of $\mathbb{F}_p$: there is no $i \neq 0$ with $a_i = 0$, and the product of the nonzero values is -1 (Wilson). The case $p = 2$ is impossible ($a_0 \equiv a_1$).

Hint 2. Let $g = \gcd(k-1, p-1)$ and $m = (p-1)/g$. As i runs over $\mathbb{F}_p^*$, i^{k-1} runs over the subgroup H of m-th roots of unity, each value g times. $i^{k-1} = -1$ must be impossible, so $-1 \notin H$, i.e. m is odd.

Hint 3. Wilson gives $\prod_{i \neq 0}(i^{k-1}+1) \equiv 1$. From $\prod_{h \in H}(x-h) = x^m - 1$ at $x = -1$ with m odd, $\prod_{h \in H}(1+h) = 2$, so $2^g \equiv 1 \pmod{p}$. Since $g \mid k-1$, $2^{k-1} \equiv 1$ and $a_2 = 2 \cdot 2^{k-1} + 2 \equiv 4$. Conversely $k = 1$ works for odd p. Answer: only $4 \bmod p$ (that is, remainder 4 for $p \geqslant 5$, remainder 1 for $p = 3$; no k works for $p = 2$).

Problem 5

Hint 1. Shift the a_i to be non-negative. Consider the set I of polynomials $\sum_j b_j x^j \in \mathbb{C}[x]$ with $\sum_j b_j f(k + j\ell) = 0$ for all k and all $\ell \neq 0$. It is an ideal (closed under multiplication by x: replace k by $k + \ell$), it contains $R(x) = \sum_i x^{a_i}$, so $I = (Q)$ with $Q \neq 0$.

Hint 2. If $P(x) \in I$ then $P(x^m) \in I$ for every $m \geqslant 1$ (replace ℓ by $m\ell$). Hence $Q(x) \mid Q(x^m)$, and the set of roots of Q is closed under $c \mapsto c^m$.

Hint 3. A finite set closed under all powers consists of 0 and roots of unity; and if Q has any root $c \neq 0$, then $c^{\operatorname{ord} c} = 1$ is also a root, so $Q(1) = 0$. But $Q \mid R$ and $R(1) = n \neq 0$. A root 0 is impossible too ($R(0) \neq 0$ after shifting so that $\min a_i = 0$). So Q is a nonzero constant, which means $f(k) = 0$ for all k.